

VTOL Forward-Flight Aerodynamics: A Steady RANS Study of a Fixed-Wing VTOL UAV in Cruise

Serge Haddad
Department of Mechanical Engineering
American University of Beirut
sph02@mail.aub.edu

Abstract

A full-airframe 3D external aerodynamics CFD simulation of a fixed-wing VTOL UAV in forward flight is presented, carried out in OpenFOAM with a steady $k-\omega$ SST RANS model on a 7 million cells mesh. The document goes over mesh generation and y^+ verification, surface-pressure and force integration, wake and tip-vortex visualisation, and convergence assessment. At cruise ($U_\infty = 25$ m/s, $\rho = 0.9$ kg/m³, $Re_c \approx 2.2 \times 10^5$) the aircraft develops a lift coefficient $C_L = 0.567$ and a drag coefficient $C_D = 0.0494$, giving a lift-to-drag ratio of 11.5 and about 37 N of lift.

Index Terms

CFD, RANS, OpenFOAM, VTOL UAV, external aerodynamics, $k-\omega$ SST, wing-tip vortices.

NOMENCLATURE

U_∞	freestream velocity, 25 m/s
ρ	air density, 0.9 kg/m ³
ν	kinematic viscosity, 1.5×10^{-5} m ² /s
q_∞	freestream dynamic pressure, Pa
b, c	wing span (1.8 m), chord (0.13 m)
S	planform reference area, 0.234 m ²
Re_c	chord Reynolds number
C_L, C_D, C_m	lift, drag, pitching-moment coefficients
L, D	lift and drag force, N
y^+	dimensionless wall distance
k, ω	turbulent kinetic energy; specific dissipation rate

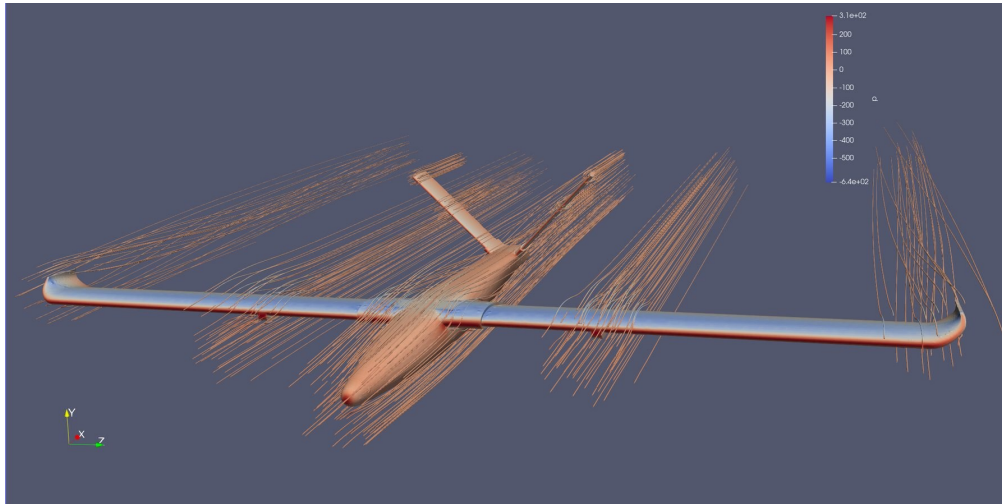


Fig. 1. Streamlines traced around the airframe in forward flight, coloured by static pressure.

I. THE CASE

The aircraft in this simulation is a VTOL with a fixed, high-aspect-ratio wing with upturned winglets, a V-tail carried on twin booms, and dedicated lift-rotor arms. This study simulates the forward-flight (cruise) condition, in which the lift rotors are idle and the wing does the work, so the model is solved as a clean external-aerodynamics problem in 3D around the full airframe (Fig. 1).

The freestream dynamic pressure and the chord Reynolds number follow from the cruise condition as

$$q_\infty = \frac{1}{2} \rho U_\infty^2 = 281 \text{ Pa}, \quad (1)$$

$$Re_c = \frac{U_\infty c}{\nu} \approx 2.2 \times 10^5, \quad (2)$$

placing the flow in a turbulent, low-Reynolds regime typical of small UAVs. The full case definition is summarised in Table I.

TABLE I
CASE SUMMARY.

Solver	foamRun: incompressibleFluid, steady-state SIMPLE
Turbulence	RANS $k-\omega$ SST with blended wall functions (high- Re near-wall treatment)
Regime	incompressible cruise; $U_\infty = 25 \text{ m/s}$; air $\rho = 0.9 \text{ kg/m}^3$
Reference	span $b = 1.8 \text{ m}$; chord $c = 0.13 \text{ m}$; planform area $S = 0.234 \text{ m}^2$
Reynolds	$Re_c \approx 2.2 \times 10^5$ ($\nu = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$) — turbulent regime
Mesh	snappyHexMesh; hex-dominant; $\approx 7\text{M}$ cells; no inflation layers
Compute	6-way MPI decomposition; 1500 iterations; $\approx 4.0 \text{ h}$ wall-clock

II. COMPUTATIONAL METHODOLOGY

A. Mesh and Near-Wall Resolution

A hex-dominant volume mesh wraps the airframe, with the surface triangulated finely over regions of high curvature (Figs. 2 and 3). There are no inflation layers: the near-wall region is captured by the snapped background cells and a graded refinement zone (Fig. 4), with wall functions bridging to the surface.

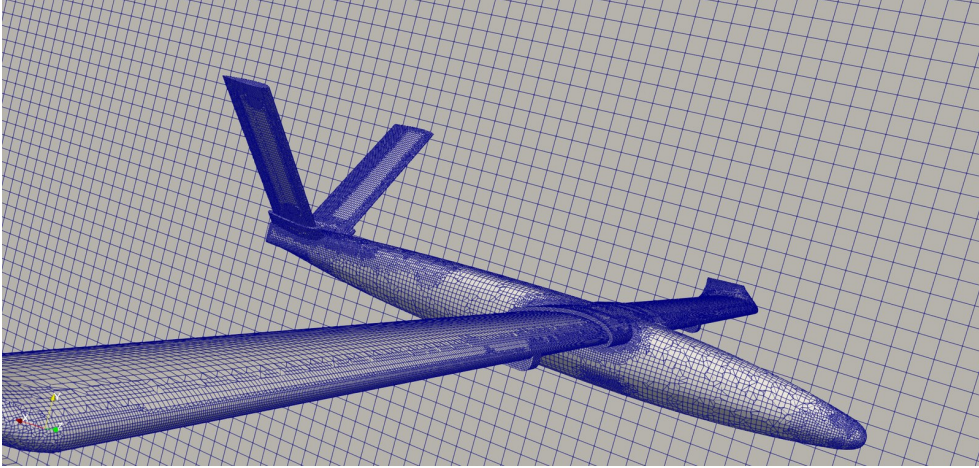


Fig. 2. Surface mesh over the full airframe.

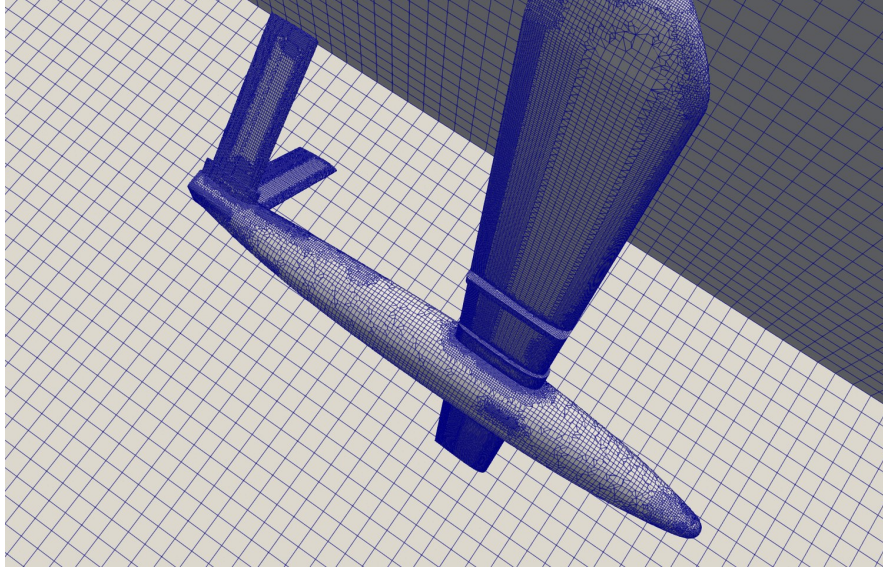


Fig. 3. Surface mesh: boom and fin detail.

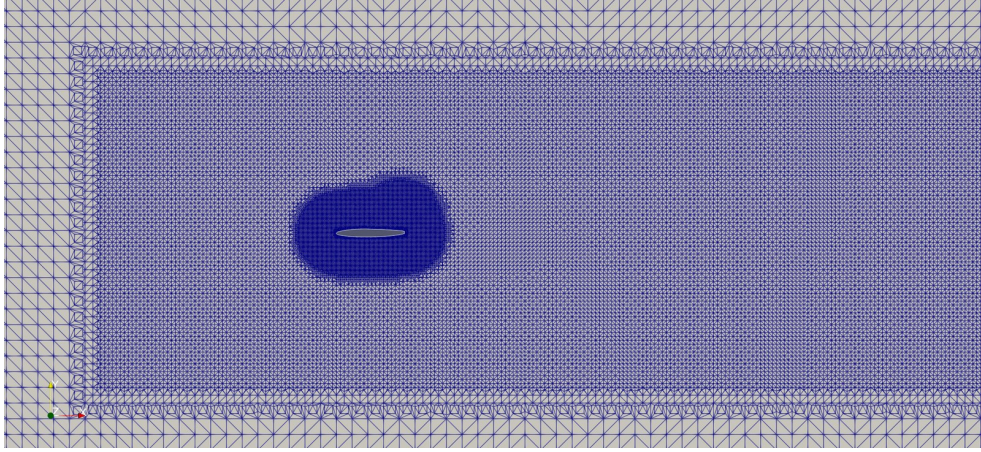


Fig. 4. Refinement regions: near-field region concentrates cells around the aircraft using snappyHexMesh distance method while the refinement box serves as an intermediary between the near-field and the coarser freestream.

B. Numerical Setup

Momentum is discretised with a second-order bounded upwind scheme for accuracy, the turbulence transport with first-order upwind for stability, and the pressure–velocity coupling uses SIMPLEC to permit aggressive under-relaxation. The governing dictionaries are listed in Table II.

TABLE II
BOUNDARY CONDITIONS AND SOLVER DICTIONARIES.

Boundary conditions	
Inlet	$U = (25\ 0\ 0)$ m/s fixed; k, ω fixed; p zeroGradient
Outlet	$p = 0$ fixedValue; U, k, ω inletOutlet
Walls	no-slip; wall functions on k, ω, ν_t ; p zeroGradient
Freestream	$k = 0.24\ \text{m}^2/\text{s}^2$; $\omega = 1.78\ \text{s}^{-1}$
Discretisation (fvSchemes)	
Time	steadyState
Gradient	Gauss linear; cell-limited on ∇U
Divergence	bounded linearUpwindV (U); bounded upwind (k, ω)
Laplacian	Gauss linear corrected
Coupling and solvers (fvSolution)	
Algorithm	SIMPLEC (consistent); 0 non-orthogonal correctors
Pressure	GAMG + Gauss–Seidel, tol 10^{-7}
$U/k/\omega$	smoothSolver (Gauss–Seidel)
Relaxation	$U = 0.9$; $k, \omega = 0.5$

C. Wall Resolution (y^+)

With wall functions, the near-wall cells must sit in the log-law band, $y^+ \gtrsim 30$. Mapped over the airframe (Figs. 5 and 6), y^+ sits squarely in that range, rising only along mid-chord acceleration lines, so the wall-function assumption holds across the wetted surface.

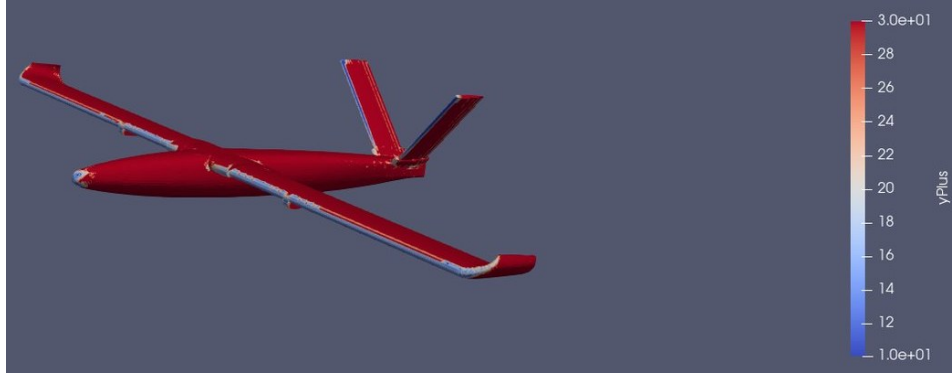


Fig. 5. y^+ clamped to 10–30: the surface is almost entirely ≥ 30 .

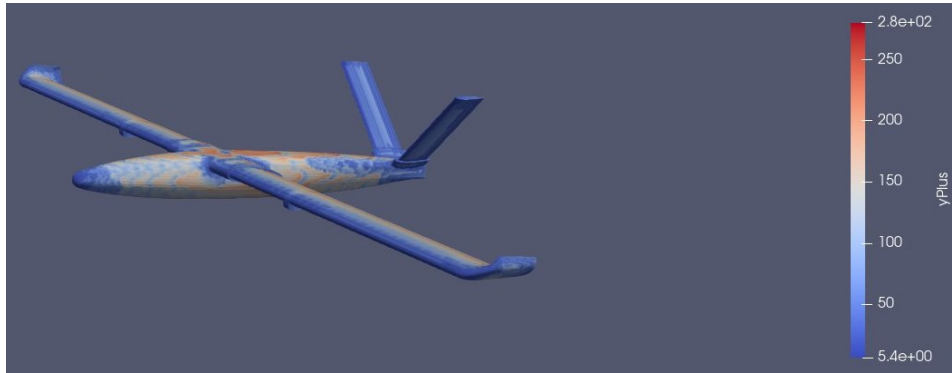


Fig. 6. y^+ over the airframe, full range 5.4–280.

III. RESULTS

The converged aerodynamic coefficients are collected in Table III; the surface-pressure field and the wake are discussed in turn below.

TABLE III
KEY AERODYNAMIC RESULTS.

C_L	0.567	C_m	1.38
C_D	0.0494	$C_{L,front}$	1.66
L/D	11.5	$C_{L,rear}$	-1.10
L	37.3 N	D	3.3 N

A. Surface Pressure and Loads

Static pressure on the skin (Figs. 7 and 8) shows strong suction over the wing upper surface generating lift, with stagnation at the nose, winglet and boom leading edges. The force coefficients are defined as

$$C_L = \frac{L}{q_\infty S}, \quad C_D = \frac{D}{q_\infty S}, \quad (3)$$

and integrating the surface tractions yields $C_L = 0.567$ and $C_D = 0.0494$ (Table III). Re-dimensionalising through (3),

$$L = C_L q_\infty S \approx 37.3 \text{ N } (\approx 3.8 \text{ kgf}), \quad D \approx 3.3 \text{ N}, \quad (4)$$

so the airframe trims out roughly a 3.8 kg cruise weight at this speed, at $L/D \approx 11.5$.

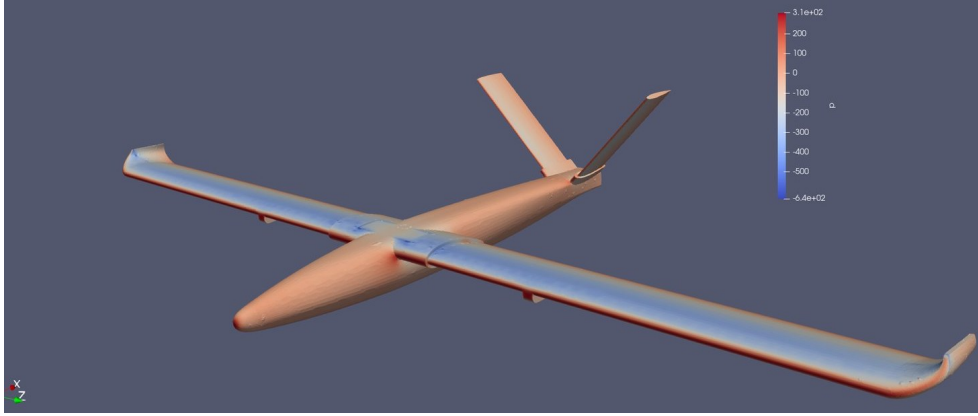


Fig. 7. Static pressure: upper-surface suction over the wing.

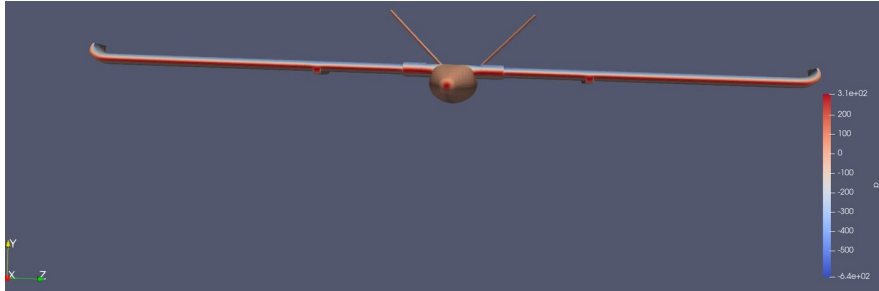


Fig. 8. Static pressure, head-on: stagnation at the nose, winglet and boom leading edges.

B. Wake and Wing-Tip Vortices

Streamlines seeded ahead of the aircraft and traced downstream reveal the wake structure (Fig. 9). At the wing tips the pressure difference rolls the flow up into a coherent tip vortex, which is the signature of finite-wing induced drag. (Figs. 10 and 11).

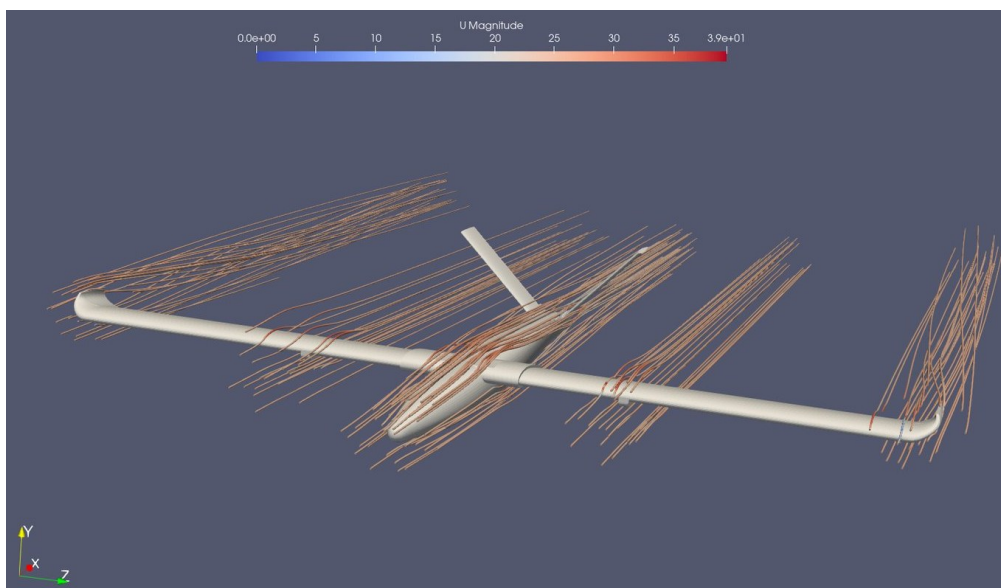


Fig. 9. Streamtubes around the whole aircraft, coloured by velocity magnitude (0–39 m/s).

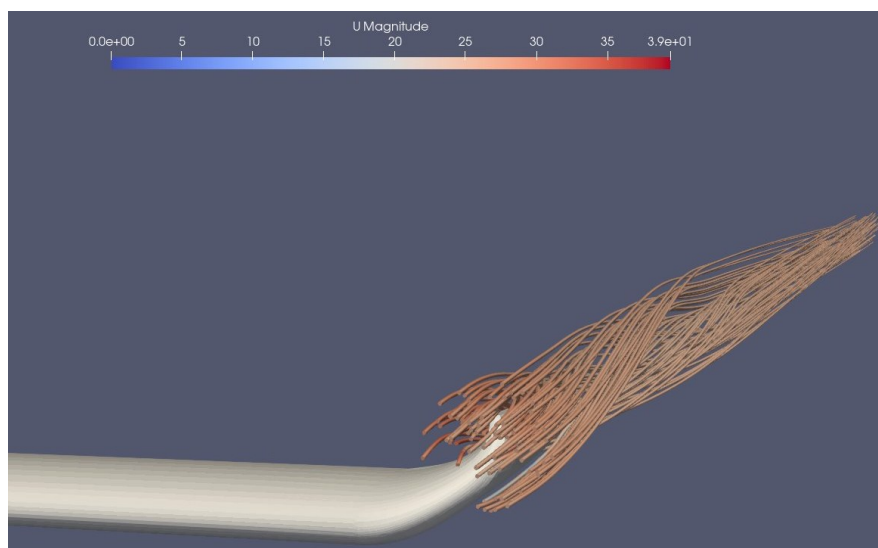


Fig. 10. Tip-vortex roll-up off the winglet, traced as streamtubes.

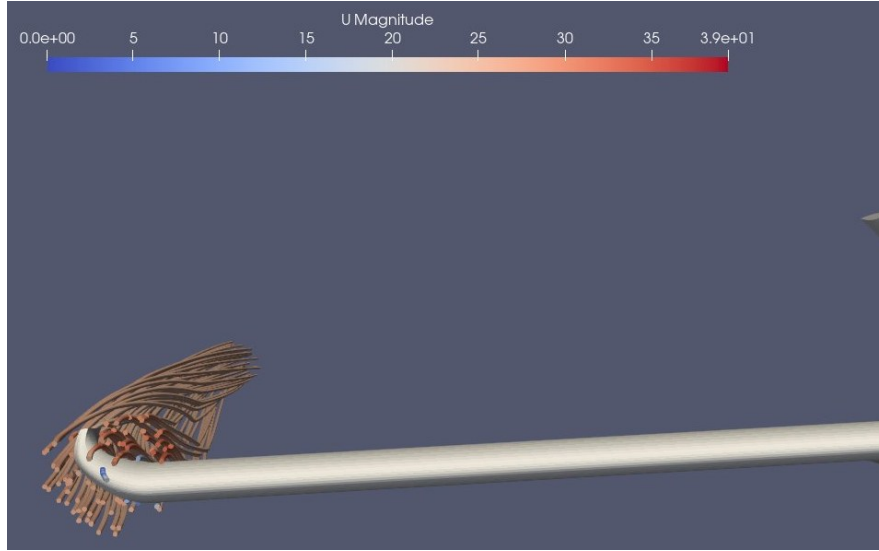


Fig. 11. Tip vortex, side-on: the rotating core convecting downstream.

IV. CONVERGENCE

The residual and force histories, extracted from the solver log, are shown in Figs. 12–14. Momentum (U_x, U_y, U_z) and turbulence (k, ω) residuals reach 10^{-4} – 10^{-7} ; the pressure residual holds around 10^{-3} at steady state, and the force coefficients flatten onto their reported values, confirming a converged solution rather than a drifting one.

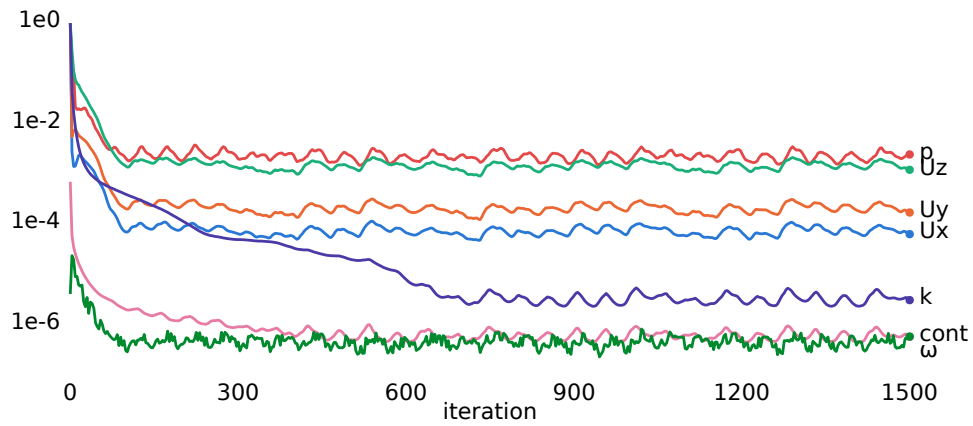


Fig. 12. Residual history (log scale): p , continuity, U_x, U_y, U_z, k, ω .

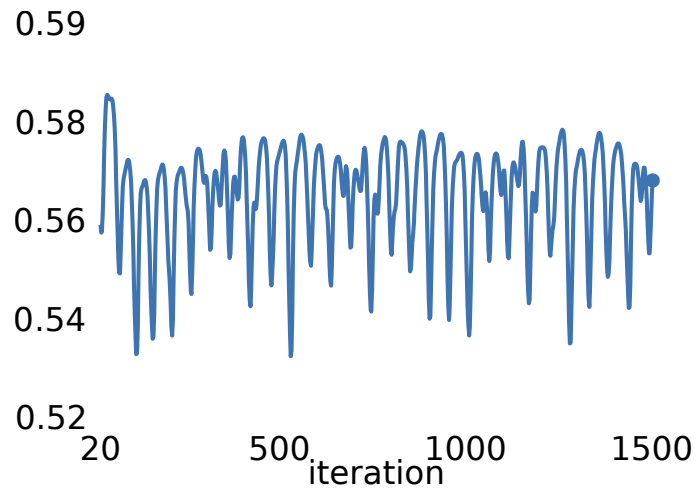


Fig. 13. Lift-coefficient history; C_L settles to 0.567 (shown from iteration 20).

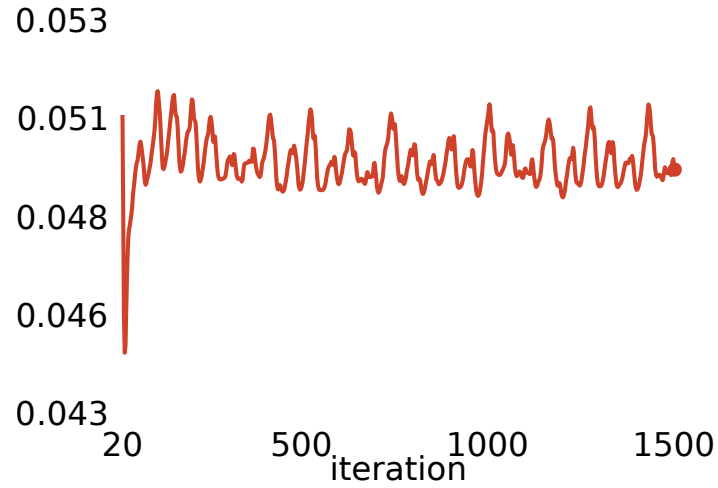


Fig. 14. Drag-coefficient history; C_D settles to 0.0494 (shown from iteration 20).

V. CONCLUSION

A converged, full-airframe RANS solution of the VTOL UAV in cruise gives $C_L = 0.567$, $C_D = 0.0494$ and $L/D \approx 11.5$, with a well-resolved boundary layer (y^+ in the log-law band), physically consistent surface loads, and clean wing-tip vortex roll-up. The workflow: meshing, solver setup, y^+ and convergence verification, and post-processing, is presented end to end. While the results don't show any irregularities, this study is only to be used as an educated engineering guess to design our scaled down VTOL prototype, pending a mesh independence study and case validation.